

Leveraging Large Language Models for Structured Data Extraction from Conversation

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Abstract — Large language models show substantial zero-shot information-extraction (IE) capability, yet conversational text — informal, multi-turn, and context-dependent — remains poorly served by IE systems designed for static, formally structured documents. This paper reports an experimental study into adapting an open-source 8-billion-parameter model, Qwen3-8B, for structured extraction from clinical doctor-patient dialogue, using a custom 798-example benchmark built from the MTS-Dialog dataset. Four research questions are examined: whether parameter-efficient LoRA fine-tuning outperforms zero- and few-shot prompting (RQ1); whether explicit annotation guidelines embedded in the fine-tuning prompt improve generalisation to unseen clinical specialties (RQ2); which of three structured output representations minimises hallucination (RQ3); and how extraction performance degrades with conversational complexity (RQ4). Naive fine-tuning on 250 examples gave null results relative to prompting (macro-F1 0.757 vs. 0.776 few-shot), whereas guideline-conditioned fine-tuning yielded a modest, real improvement (0.780) with reduced hallucination. JSON output minimised hallucination relative to natural-language and Python-dataclass methods, and a measurable but partly artefactual generalisation gap was observed between seen and unseen specialties. Following supervisor-approved scope narrowing to a single clinical domain, the resulting benchmark, evaluation harness, and findings are presented as a foundation for future development.

Keywords – *Large Language Models; Information Extraction; LoRA Fine-Tuning; conversational AI; Clinical Natural Language Processing.*

I. INTRODUCTION

A. Problem and Context

Information extraction (IE) is the transformation of unstructured text into structured, machine-readable records; it has traditionally assumed static, well-formed input: news articles, scientific abstracts, legal filings. Conversational data violates every one of these assumptions. Statements are informal, incomplete, spread across multiple turns between speakers and mostly rely on implicit context (e.g. a symptom mentioned three turns earlier referenced only as “it”). This gap matters more as organisations accumulate large volumes of conversational data — clinical encounters, customer-service transcripts and voice-assistant logs — that must be mined for structured, actionable content.

Large language models have demonstrated substantial zero-shot IE capability, but this capability is neither free nor uniform. GPT-3.5 achieves only 18.22 F1 on standard named-entity recognition without task-specific adaptation [3], and zero-shot dialogue-state tracking trails fine-tuned models by 10–20 points on MultiWOZ [6], a benchmark built on the neural dialogue-systems paradigm established by earlier pretrained-language-model approaches [5]. Existing benchmarks rarely assess entity, relation, intent, and event extraction jointly within multi-turn conversational context, and this gap is sharpest in high-risk domains such as healthcare, where extraction errors carry real consequences.

B. Research Aim and Objectives

The aim of this project is to determine how an open-source LLM can be accurately adapted, through fine-tuning, prompt design, and output-format selection, to extract structured clinical facts from doctor-patient conversation, and to characterise the conditions under which each adaptation method succeeds or fails. This splits into four objectives, each mapped to one research question addressed in Section IV:

- I. Establish whether parameter-efficient fine tuning improves on prompting alone.
- II. Establish whether explicit annotation guidelines improve generalisation to specialties unseen during training
- III. Establish which structured-output representation best balances accuracy against hallucination.
- IV. Characterise how extraction quality becomes worse with conversational complexity.

The expected contributions are threefold: a conversational IE benchmark for the clinical domain; a reusable system for scoring three output formats against identical gold labels; and an honest empirical account of where small-scale fine-tuning performs both above and below prompting.

C. Scope and Boundaries

Following a supervisor-approved scope review, the study is confined to healthcare conversations within the MTS-Dialog corpus and does not extend

to a multi-domain benchmark, proprietary-model baselines (GPT-3.5/4), or reproduction of the baseline approaches (InstructUIE, GoLLIE, FnCTOD) originally envisaged. A fifth research question, on joint multi-format generalisation, was also no longer considered, in line with the project's risk-mitigation plan. As a direct consequence of this narrowed scope, the findings reported here should be read as specific to a single clinical domain rather than as general claims about conversational information extraction.

D. Research Questions

RQ1: Does parameter -efficient fine tuning (LoRA) outperform zero-shot and few-shot prompting for conversational information extraction?

RQ2: Does embedding explicit annotation guidelines in the fine-tuning prompt improve zero-shot generalisation to clinical specialties unseen during training?

RQ3: Does a JSON function-calling output representation reduce hallucination and improve structural validity relative to natural-language labels and python data class instances?

RQ4: How does extraction performance degrade across levels of conversational complexity and between seen and unseen specialties?

E. Methodology Overview and Paper Organisation.

An experimental machine-learning methodology is followed across four phases: benchmark creation, model adaptation, evaluation and error analysis. The key literature behind this methodology is outlined in Section 2. Details about the data set model architecture setup, and the outcomes of RQ1 to RQ 4 are described in Section 3. The conclusions and limitations of this research are covered in Section 5.

II. LITERATURE REVIEW

Four papers form the core literature basis for this project, jointly informing its treatment of fine-tuning versus prompting, guideline-conditioned training, and structured-output-format design, though none combines all three within a conversational, clinical setting.

On-Demand Information Extraction (ODIE) [1] fine-tunes LLaMA-7B with LoRA on a ChatGPT-generated instruction-extraction dataset, pulling user-specified content into structured tables from free text. ODIE outperforms its baseline by nearly 5% and approaches GPT-4 despite being far smaller, showing that a fine-tuned open model can compete with a proprietary one on structured extraction. Its six-step data-generation pipeline is the direct template this project adapts for benchmark construction. Its limitation is that it operates only on

static documents, leaving multi-turn context resolution and speaker coreference unaddressed.

Zero-shot dialogue-state tracking via function calling (FnCTOD) [2] represents each dialogue domain as a callable function whose arguments are the user-specified slot values; a fine-tuned FnCTOD-LLaMA2-13B matches ChatGPT's tracking performance. Its JSON function-call format is the direct precedent for this project's JSON output condition (RQ3). Its limitation is a closed, fixed slot schema that cannot express the open-ended free-text fields clinical extraction requires.

InstructUIE [3] unifies named-entity, relation, and event extraction under a single instruction-tuned FlanT5-11B model and generalises to seven domains unseen during its own training, motivating this project's seen/unseen specialty split (RQ4). Its natural-language output format is the basis for this project's "NL" condition. It reports no hallucination metric; this project's RQ3 finding that the NL-label format hallucinates far more (33.8–58.0%) than a JSON alternative (7.5–9.7%) directly extends its evaluation.

GoLLIE [4] trains models to follow explicit annotation guidelines, expressed as Python dataclasses, combined with five regularisation techniques, allowing a 7B model to outperform an 11B InstructUIE by 13 F1 points with under 1% hallucination. It is the direct source of this project's dataclass output condition and the entire premise of RQ2. Its limitation relative to this project is that it is evaluated only on static news and biomedical text, using substantially more regularisation than is reproduced here.

Read together, these papers reveal a clear gap: extraction and generalisation are studied without dialogue (ODIE, InstructUIE); dialogue is addressed only within closed, task-oriented schemas (FnCTOD); and guideline-conditioning with output-format hallucination control is demonstrated only on static text (GoLLIE). No existing work combines a fine-tuning-versus-prompting comparison, guideline conditioning, and output-format control within a single conversational, open-ended-field benchmark. This project's contribution is precisely that combination, evaluated in the clinical domain under identical conditions on identical held-out data.

III. METHODS

A. Dataset

The benchmark was built from 800 conversations sampled from the MTS-Dialog dataset, a publicly available collection of doctor-patient dialogue transcripts developed for clinical note-generation research. Each conversation was passed through Qwen3-8B to extract five fields against a fixed

schema: symptoms (name/duration/severity), family_history (relation/condition), medications (name/dose/frequency), diagnosis (free text), and recommendation (free text) — a mix of structured list fields and open free-text fields, bridging FnCTOD’s closed slot-value schema and ODIE’s open table extraction.

The annotation prompt includes direct handling for two frequent error types identified during early trial runs. Negation: a symptom the patient clearly denies (e.g. “no, I haven’t”) must not be extracted as present, since a naive extractor otherwise records the symptom regardless of the surrounding denial. Attribution: a fact belonging to a family member or another person mentioned in the conversation must not be recorded as the patient’s own symptom, medication, or diagnosis. Both rules were added after manual inspection showed each error type occurring systematically rather than incidentally, and both are checked again, independently, at the self-verification stage described below.

Two automatic quality filtering methods were tested on the raw annotations before human review. An NLI based regularity check, testing whether each extracted fact was entailed by the source text, was implemented and tested but abandoned: it proved unreliable on long, multi turn medical dialogue, flagging 717 of 1,687 facts, many of which were correct on inspection. It was replaced with Qwen3 self-verification pass, and the same model family fact-checks its own extraction against the source conversation and the whole human-written summary, flagging 275 facts (16.3%) for review. A full human review pass then corrected every flagged fact, most commonly negation errors and one misattribution case.

TABLE 1. COMPOSITION OF THE LOCKED BENCHMARK (798 EXAMPLES)

Dimension	Breakdown
Total examples	798
Specialty split	Seen: 709 (88.8%) / Unseen: 89 (11.2%)
Difficulty tier	Easy: 299 / Medium: 257 / Hard: 242
Held-out specialties	Dermatology, Orthopedics

The 88.8/11.2 seen/unseen split and the two held out specialties were fixed before any fine-tuning experiment and reused, unchanged, across RQ1, RQ2, and RQ4, so generalisation results are directly comparable across research questions. The 13 sub-specialties are themselves unevenly represented (Table 2).

TABLE 2. SUB-SPECIALTY DISTRIBUTION ACROSS THE 798-EXAMPLE BENCHMARK

Sub-specialty	n
General medicine	266
Cardiology	104
ENT	94

Gastroenterology	60
Orthopedics (unseen)	59
Neurology	51
Endocrinology	35
Dermatology	30
Pulmonology	26
Mental health	22
Gynecology	18
Oncology	17
Urology	16

B. Model Architecture

The backbone model is Qwen3-8B, an open-source, instruction-tuned, decoder-only transformer, selected for strong reported instruction-following at the 7–8B scale, open weights permitting full gradient access unavailable with GPT-3.5/4, and mature LoRA/PEFT tooling. The model is loaded in 4-bit precision, a choice made necessary by the 8 GB VRAM laptop GPU. A LoRA adapter is attached to the four attention projection matrices (query, key, value, output) with rank 16, scaling alpha 32, and dropout 0.05, training 15,335,424 of 8,206,070,784 parameters (0.187%) while leaving base weights frozen. Rank, alpha, and target-module selection follow validated LoRA and GoLLIE methods rather than a project-specific search; a relatively low rank was preferred over a larger value specifically because the RQ1/RQ2 training subsample contains only 250 examples, where a higher-capacity adapter would carry a greater risk of overfitting at the cost of some representational flexibility.

C. Training Procedure

Training data for the RQ1 condition consists of 250 of 709 seen examples (fixed seed = 42), each pairing the extraction prompt with the JSON annotation as the target completion; prompt tokens are masked so cross-entropy loss is computed only over the target completion. Training uses a per-device batch size of 1 with gradient accumulation over 8 steps, one epoch, learning rate $2e-4$, fp16 precision, and the paged 8-bit Adam optimiser, with gradient checkpointing enabled. For RQ2, an otherwise identical script substitutes an alternative, guideline-augmented prompt — the same JSON schema prefaced with per-field inclusion/exclusion rules — while every other hyperparameter is held fixed, isolating guideline presence as the sole independent variable.

D. Hyperparameter Rationale and its Limits

Hyperparameters were derived from established literature rather than a project-specific search, a direct consequence of the single-consumer-GPU compute budget: each fine-tuning and evaluation cycle required over an hour of GPU time, making a multi-point hyperparameter sweep impractical within the available timeframe. Two engineering issues were identified and fixed during the RQ2 run.

First, longer guideline-augmented prompts caused 71% (177/250) of training examples to exceed the original 640-token cap, silently truncating the target completion; this was corrected by compacting the guideline text and raising the cap to 768 tokens, reducing the truncation rate to 8% (20/250). Second, GPU memory pressure at 96–97% of the 8 GB card caused Windows to spill allocations into system RAM; switching gradient checkpointing to non-reentrant mode gave partial relief. These constraints are treated as limitations rather than settled methodological choices.

IV. EXPERIMENTATION AND RESULTS

A. Hardware and Software Configuration

All experiments were run on a single local machine, following an initial move away from Google Colab, whose shared GPU allocation proved unreliable for multi-hour fine-tuning runs. The configuration is listed in the table below to support independent reproduction.

TABLE 3. HARDWARE AND SOFTWARE CONFIGURATION FOR ALL REPORTED EXPERIMENTS

Component	Specification
GPU	NVIDIA RTX 5060 Laptop GPU (8 GB VRAM)
OS	Windows 11
PyTorch	2.11.0+cu128 (Blackwell sm 120 kernel support)
Transformers	5.15.0
PEFT	0.20.0
bitsandbytes	0.50.0
Base model	Qwen / Qwen3-8B (Hugging Face)

Both were verified via an explicit GPU matrix-multiplication test, not merely a version check, before any experiment was trusted to have run on GPU.

B. Evaluation Metrics and Rationale

The plan’s originally specified strict-span F1 metric was judged after inspection, poorly suited to free-form clinical text, where a correct extraction may take many surface forms (eg. “Kids” vs “Children”). Two adjusted metrics were used instead, applied identically across every RQ: fuzzy slot-level F1 (list fields: token-Jaccard overlap at a 0.5 threshold, so paraphrases count as a match) and token-overlap F1 (free-text fields: precision/recall over predicted vs gold token sets). Macro-F1, the unweighted mean of the five per-field scores, is used as the single headline number. Hallucination rate (the fraction of extracted facts whose tokens do not appear in the source conversation) and structural validity (parse success rate) are reported alongside macro-F1 throughout, consistent with the plan’s focus on grounding and format reliability.

C. RQ1 Results: Fine-Tuning vs Zero/Few-Shot Prompting

TABLE 4. BASELINE QWEN3-8B ZERO-SHOT/ FEW-SHOT MACRO-F1 ALL 798 ITEMS

Field	Zero-shot F1	Few-shot F1
Symptoms	0.8171	0.8091
Family history	0.8339	0.8317
Medications	0.9632	0.9571
Diagnosis	0.7429	0.8560
Recommendation	0.7563	0.8165
Macro average	0.8227	0.8541

Parse failures were minimal for both baseline conditions (3/798, 0.4% zero-shot; 1/798, 0.1% few-shot), confirming that the baseline comparison in Table 4 is not confounded by a meaningful difference in output-format reliability between conditions.

TABLE 5. RQ1: ZERO-SHOT VS FEW-SHOT VS LORA FINE-TUNED UNSEEN SPLIT (N=89)

Condition	n	Macro-F1
Zero-shot	89	0.756
Few-shot	89	0.776
Fine-tuned (LoRA, no guidelines)	89	0.757

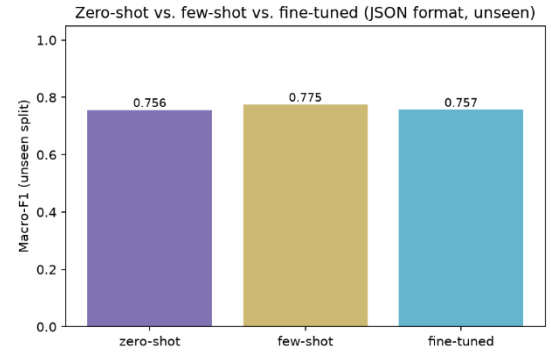


Fig. 1. Zero-shot vs few-Shot vs fine-tuned macro-F1 on the unseen split (RQ1)

Fine-tuning essentially matched zero-shot prompting (+0.001 F1) and underperformed few-shot prompting (−0.019 F1) on the unseen split — a genuine null result relative to the plan’s anticipated 5–15-point improvement, reported as such rather than reframed.

D. RQ2 Results: Annotation-Guideline Ablation

TABLE 6. RQ2: GUIDELINE-CONDITIONED VS NON-GUIDELINE FINE-TUNING, UNSEEN SPLIT (N=89)

Condition	n	Macro-F1	Hallucination
Zero-shot	89	0.756	—
Few-shot	89	0.776	—
Fine-tuned, no guidelines	89	0.757	12.8%
Fine-tuned, with guidelines	89	0.780	11.2%

Including explicit rules for annotation in the fine-tuning prompt led to gains of +0.023 in macro-F1 and −1.5 in hallucination in the unseen

split—the highest-performing setting seen for both RQ1 and RQ2, but somewhat lower than the reported 8-13 point gain for GoLLIE in its own, more regulated setting.

E. RQ3 Results: Output-Format Comparison

TABLE 7. RQ3: OUTPUT-FORMAT COMPARISON ACROSS ALL 798 ITEMS

Format / mode	Struct. valid.	Halluc. rate	Macro-F1
JSON, zero-shot	99.6%	9.7%	0.823
JSON, few-shot	99.9%	7.5%	0.854
NL labels, zero-shot	99.7%	33.8%	0.592
NL labels, few-shot	100.0%	58.0%	0.429
Dataclass, zero-shot	98.2%	12.1%	0.747
Dataclass, few-shot	98.6%	26.1%	0.680

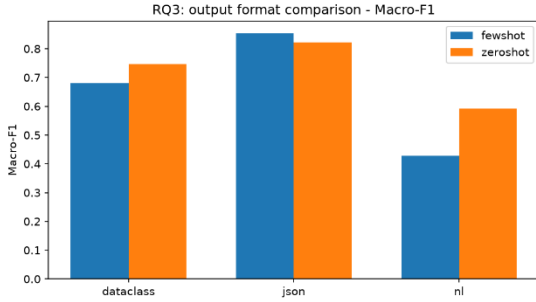


Fig. 2. RQ3 macro-F1 by output format and prompting mode

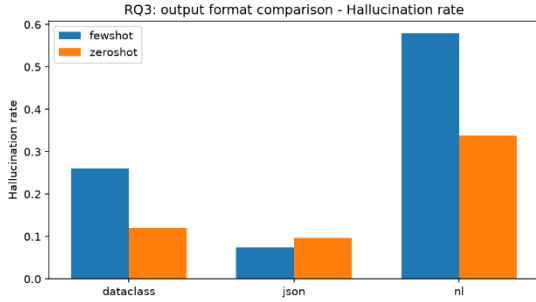


Fig. 3. RQ3 hallucination rate by output format and prompting mode

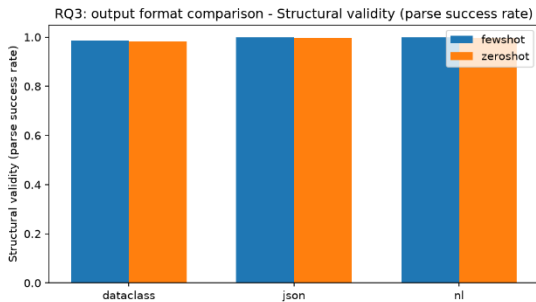


Fig. 4. RQ3 Structural validity (parse success rate) by output format and prompting mode

JSON is the clear winner on every criterion — maximum structural validity, minimum hallucination, and maximum macro-F1 for both prompting setups — precisely in line with the intended outcome of the study design. An additional finding is that few-shot prompts increase hallucination for both the NL-label and dataclass formats but decrease it for JSON: the more confined generation space of JSON appears to let few-shot examples serve grounding rather than pattern completion.

F. RQ4 Results: Complexity-Tier and Generalisation Degradation

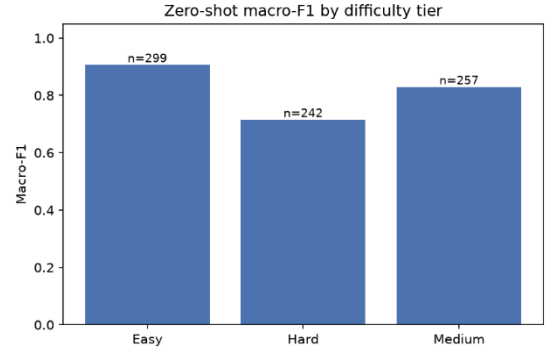


Fig. 5. Zero-shot macro-F1 by conversational difficulty tier.

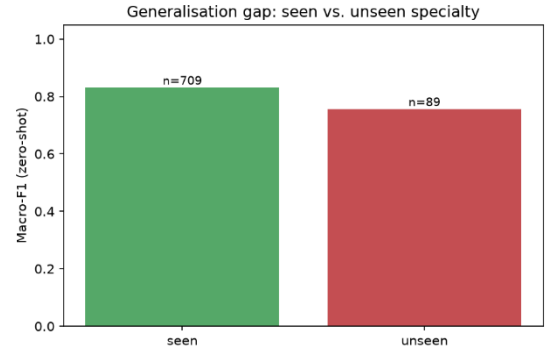


Fig. 6. Generalisation gap: zero-shot macro-F1, seen vs unseen specialty

Seen specialty zero-shot macro-F1 score (0.831) is 7.5 percentage points higher than that for unseen specialty macro-F1 (0.756). The 30 worst-scoring instances in the unseen category, 37% turned out to be cases of artefactual scoring (ambiguity of blank gold label and pluralisation issue), 50% were actual extraction failures, while 13% were mixed cases. It can thus be concluded that the actual generalization gap is smaller than the raw number implies, yet there is still one present.

G. Supplementary Results: Model-Size Comparison

TABLE 8. QWEN1-1.7B VS QWEN3-8B, ALL 798 ITEMS

Model	Zero-shot F1	Few-shot F1
Qwen3-1.7B	0.772	0.773
Qwen3-8B	0.823	0.854

This additional experiment, conducted by way of an inexpensive enhancement to the current harness for performance evaluation, measures the cost of a model being about 4.7x smaller in terms of F1 macro-reduction: 5.1 points (zero-shot) or 8.1 points (few-shot).

H. Comparison Against Anticipated Results

TABLE 9. ANTICIPATED VS ACTUAL RESULTS RQ1-RQ3

RQ	Anticipated	Actual
RQ1	+5 to +15 F1 pts	+0.001 pt (null result)
RQ2	+8 to +13 F1 pts	+2.3 pts (real, modest)
RQ3	JSON: lowest halluc., highest validity	Confirmed exactly

The answer to RQ3 is not unexpected due to its narrow scope of generation space. The deficiency in meeting the expectations of RQ1 and RQ2 is critically evaluated in Section V; the primary theory behind this is that having only 250 instances in a single epoch does not adequately train the adapter.

Table 10 situates this project’s RQ3 hallucination rates against the two literature baselines that report a comparable metric.

TABLE 10. HALLUCINATION RATE AGAINST LITERATURE BASELINES (STATIC TEXT) VS. THIS PROJECT (CONVERSATIONAL TEXT)

System / Condition	Output Format	Hallucination Rate
GoLLIE [4] (static text)	Python dataclass + 5-technique regularisation	< 1%
InstructUIE [3] (static text)	Natural-language labels	Not reported
This project (conversational)	JSON function-calling	7.5–9.7%
This project (conversational)	Python dataclass	12.1–26.1%
This project (conversational)	Natural-language labels	33.8–58.0%

GoLLIE’s near-zero hallucination rate reflects both its five-technique regularisation regime and its static-text setting; this project’s JSON condition, evaluated on multi-turn conversational text with no comparable regularisation, hallucinates substantially more but still far less than either alternative output format tested here, reinforcing JSON as the appropriate default for hallucination-sensitive conversational extraction.

V. DISCUSSION

A. Interpretation of Results

RQ1’s null result represents an important empirical finding, not merely a failed experiment: for a modern, highly instruction-tuned 8B model, prompting with a small number of in-context examples may already capture nearly all the performance extractable from 250 examples, and a small, single-epoch LoRA fine-tune does not easily translate that same budget into a comparable gain. This raises a practical question for a model user: whether to spend a fixed example budget on in-context examples or on fine-tuning. The results here favour the former, contrary to the implicit assumption behind the original project proposal.

RQ2’s modest, positive result is arguably the project’s most actionable finding for practitioners: explicit annotation guidelines, added at effectively zero marginal training cost — the guideline-conditioned run uses the identical 250-example subsample, LoRA configuration, and training arguments as the no-guidelines run, with only the prompt text differing — yielded both higher accuracy and lower hallucination. Although the magnitude (2.3 points) falls short of GoLLIE’s own reported 13-point gain, the direction and consistency of the effect (simultaneously higher F1 and lower hallucination, not a trade-off between the two) suggests guideline-conditioning is a genuinely useful, low-cost lever, and that GoLLIE’s larger reported gain likely reflects its additional regularisation techniques (class-order shuffling, dropout, paraphrasing) rather than guideline text alone — a hypothesis this project’s simpler ablation is well positioned to have surfaced, precisely because it isolates guideline text as the sole variable.

RQ3, clean and unambiguous across every metric, is the project’s strongest result, and its practical implication follows directly: for a hallucination-sensitive, high-stakes domain such as clinical extraction, JSON function-calling output should be the default choice, not merely one option among three. The finding that few-shot examples increase, rather than decrease, hallucination for the NL-label and dataclass formats is a genuinely novel observation relative to the reviewed literature and merits further mechanistic investigation — for example, whether it reflects in-context examples inadvertently licensing free elaboration in less-constrained output grammars. RQ4’s finding — that roughly a third of the apparent seen/unseen generalisation gap is a scoring artefact rather than a genuine model failure — is a caution against over-interpreting any single aggregate metric without manual error analysis, and reinforces the value of the fuzzy/token-overlap metric design over a stricter alternative that would have obscured, rather than revealed, this distinction.

B. Implication and Significance

As a whole, these results imply a clear recommendation for practitioners developing conversational clinical IE systems within a limited compute budget: prioritise output-format engineering (JSON) and prompt/fine-tuning conditioning over extensive fine-tuning effort, at least until a larger training-data budget is available. This conclusion contradicts the common assumption, implied throughout the reviewed literature, that fine-tuning is the primary mechanism for domain adaptation.

C. Limitations

- **Monodomain:** all findings are domain-specific to clinical/healthcare dialogue from the MTS-Dialog dataset; generalisability to other conversational domains (e.g., customer support, e-commerce, legal intake), originally planned for in the project plan, is untested.
- **Small unseen split sample size:** the RQ1/RQ2 headline comparison relies on 89 unseen instances: an observed 1–2-point macro-F1 difference at such a sample size is not significant enough to claim conclusively without significance testing (e.g., bootstrapped confidence intervals) before being treated as conclusive, which was not done within the scope of this project.
- **Proprietary model not included:** the lack of a GPT -3.5/4 comparison due to the lack of API budget hinders the open-source vs proprietary aspect originally envisioned in the plan; all model comparisons in this paper are done internally using Qwen.
- **Not reproduced against literature baselines:** although InstructUIE, GoLLIE 7B, and FnCTOD-LLaMA2-13B are covered in detail in Section 2, none of those models were tested on this benchmark, due to hardware limitations and the descopeing of the benchmark.
- **The Annotator-Verifier correlation:** both the annotation phase and the self-verification phase employ models belonging to the same family, i.e., Qwen3; any inherent extraction tendency in such models is potentially imperceptible through self-verification as the verifier suffers from the same flaws as the annotator.
- **Search of hyperparameter space incomplete:** the selection of parameters such as LoRA rank, LoRA alpha, learning rate, and epochs was made based on literature recommendations, and therefore, the negative result for RQ1 cannot be

separated from an under-training effect entirely.

- **Human evaluation incomplete:** an evaluation package consisting of 60 questions has been developed but not yet implemented, and therefore, the automatic evaluation measures that were used throughout this paper have not yet been validated with human evaluation

VI. CONCLUSION AND FUTURE WORK CONSIDERATION

A. Summary of Contributions

This project aimed to establish how a publicly available LLM can be fine-tuned for structured extractions from clinical conversations, considering four research questions on fine-tuning, guideline conditioning, output format and complexity degradation. It presents a 798-example, human-annotated, stratified benchmark of structured extraction from clinical conversation; a reusable evaluation harness that rates three structured output formats using identical gold labels; a genuine, positive finding in relation to guideline-conditioned fine-tuning: a definitive, literature-revealing finding for JSON output format, as well as a novel finding relating to few-shot examples and hallucination; and a stratified degradation analysis that distinguishes between genuine generalization and scoring artifacts. The RQ2 adapter was also merged into a standalone model (`merge_lora_adapter.py`) powering an interactive extraction demo used in the recorded video demonstration.

B. Critical Discussion of Limitations

The most significant limitation is that of scope: with only one clinical area, no baseline from either the proprietary data or the literature, and unexplored hyperparameters, the results here can be seen as specific to the domain and the particular configuration tested and not as generalized statements about fine-tuning, guideline conditioning, or output formatting broadly. The sample size of the RQ1/RQ2 study ($n = 89$) further implies that while the point estimates provided are useful in terms of direction, they have not been subjected to statistical significance testing and then remain suggestive of, but not conclusive in, their findings.

C. Future Work

- **Validation through bootstrapped confidence intervals** of the delta on RQ1 and/or RQ2.
- **Data scale and number of epochs** ablation study to check if the negative

results on RQ1 are due to lack of training rather than true ceiling effects.

- Empirical validation in additional domains using the benchmark generation pipeline.
- At least partial proprietary model comparison.
- Complete creation of the constructed human evaluation protocol to validate the automatic evaluation metrics against human judgments.
- More comprehensive GoLLIE-style regularization scheme on RQ2 to isolate the contribution of regularisation technique from data/compute scale

D. Unresolved Questions

Two questions remain genuinely open: whether RQ1’s null result reflects a fixable data/compute constraint or a real ceiling effect in strongly instruction-tuned models, and the precise mechanism behind RQ3’s few-shot-driven increase in hallucination for less-constrained output formats.

VII. AI ETHICAL CONSIDERATION

A. Bias

The benchmark’s specialty distribution is imbalanced (Table 2): general medicine accounts for 33.3% of all examples, while the two least-represented specialties (oncology and urology) account for just 2% each. Any model evaluated against this benchmark will therefore show inconsistent performance across specialties, with those most underrepresented in the data showing the worst results. This is compounded by the fact that the initial annotation and self-verification steps use the same underlying model family: a systematic bias in how that model perceives specialty-specific vocabulary risks slipping through the verification stage undetected.

B. Privacy

This corpus refers to a publicly available research data set that does not come from real-life patients. Future application to real-life clinical data will need to be accompanied by de-identification, compliance, and institutional ethics, none of which is in scope of the current project.

C. Fairness

The seen versus unseen performance difference is a fairness issue if such a system were deployed in practice, as the patients of underrepresented medical specialties would suffer from systematically inferior quality of extraction, without letting the doctor know about it.

D. Transparency

Hallucination rate is presented as a first-class metric throughout the paper, in the spirit of transparency: even the highest performing JSON output format still suffers from hallucination at a 7.5–9.7% rate, and this rate needs to be disclosed by any system based on the current pipeline.

E. Accountability and Human Oversight

Due to the hallucination rates and generalization error between seen and unseen cases discussed above, the tool is strictly used as a drafting tool for clinicians only with mandatory human review.

VIII. REFERENCE

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